When the separate universe fails: a criterion for the squeezed bispectrum in non-attractor phases

Houston Golden[1, *]

[1] *Independent Researcher, Los Angeles, California, USA*

(Dated: September 4, 2026)

The isotropic separate universe (δN) reproduces the squeezed bispectrum of ζ in ultra-slow-roll inflation, and fails by a factor of 8/3 in a matter-dominated contraction. We derive the exact threading identity relating δN's zero-shift variable δN_c to Maldacena's ζ and state a single criterion: the two agree iff the ζ-growth-weighted mean $\langle\epsilon/c_s^2\rangle_\zeta$ vanishes, otherwise $\delta N_c = (1 - \langle\epsilon/c_s^2\rangle_\zeta/3)\,\zeta$ plus an explicit second-order correction. The linear result is exact for any history; the second-order result is exact at constant ϵ, $c_s = 1$, with a closed general equation-of-state form. We validate on four cases – matter-dominated contraction ($O(1)$ failure), ultra-slow-roll (agreement to $O(\epsilon)$), attractor slow roll (identity map), ekpyrosis (passes) – and place the result against the non-attractor consistency-relation literature. The criterion is a second, independent failure mode of the separate universe beyond the known initial-data requirement, invisible to inflationary tests because it is $O(\epsilon)$ there and $O(1)$ in contraction.

I. MOTIVATION

The squeezed limit of the primordial bispectrum is governed, in single-clock inflation, by Maldacena's consistency relation, and can equivalently be computed by the separate-universe (δN) formalism, which treats the long-wavelength mode as a local shift of the background evolution [1]. In a non-attractor phase, where $\dot\zeta \neq 0$ on super-Hubble scales, two independent difficulties arise. First, Namjoo, Firouzjahi & Sasaki showed that ultra-slow-roll (USR) inflation violates the consistency relation, $f_{\rm NL} = 5/2$, and that δN built from $N(\phi)$ alone misses this: the growing mode is an independent initial datum, requiring $N(\phi, \pi)$ to be correctly captured [3]. This initial-data requirement was extended to $P(X)$ non-attractors by Chen, Firouzjahi, Namjoo & Sasaki, who found $f_{\rm NL} = 5(1 + c_s^2)/(4c_s^2)$ and confirmed δN with momentum agrees [4]. Pajer, Schmidt & Zaldarriaga separately argued that in genuine attractors the squeezed bispectrum is an unobservable coordinate artifact of the long mode [5], a statement later sharpened with conformal Fermi coordinates by Dai, Pajer & Schmidt [6]. Cai *et al.* tracked $f_{\rm NL}$ through smooth and sharp USR transitions and found δN and in-in agree throughout [7]. Bravo, Mooij, Palma & Pradenas derived a generalized consistency relation for non-attractor backgrounds and showed the canonical single-field local $f_{\rm NL}$ is removable by a change of physical coordinates [8, 9], and Passaglia, Hu & Motohashi confirmed $\delta N(\phi, \pi)$/in-in agreement for PBH-motivated USR scenarios [10]. Separately, Artigas, Grain & Vennin, and Jackson *et al.*, identified a phase-space separate-universe breakdown localized to a finite window of scales at slow-roll-to-USR transitions, traced to a momentum/initial-data mismatch [11, 12].

Second, and logically independent, a companion theory-audit note on this program found that the isotropic separate universe applied to a matter-dominated cosmological contraction ($w = 0$, $\epsilon = 3/2$) returns $f_{\delta N} = -5$ (initial-position label) while the from-scratch in-in squeezed bispectrum of ζ is $-35/16 + (15/16)\mu^2$ [18], a monopole gap of 25/8 – an $O(1)$ discrepancy, not a small correction. A second companion note derived the exact second-order map between the two variables and showed the gap decomposes into a linear rescaling and an explicit second-order term, both proportional to ϵ [19]. Every prior test of δN/in-in agreement in the literature above is an inflationary non-attractor, where $\epsilon \ll 1$ throughout the growth of the long mode. This note asks the question directly: is there a single criterion, applicable to both inflationary and contracting non-attractors, that predicts when the two constructions agree and by how much they differ when they do not.

II. THE THREADING IDENTITY AND THE MAP

Comoving gauge, ADM variables $N = 1 + \alpha$, $N_i = \partial_i\psi + \tilde N_i$. The zero-shift threading computed by the separate universe is the fluid (normal) congruence, with local expansion K; along a fluid worldline the fully nonlinear relation between K and the local volume element gives the exact identity

$$\delta N_c(x_f) = \zeta(t_f, x_f) - \frac{1}{3}\int_{-\infty}^{t_f} \partial_i N^i\big(t, x(t)\big)\, dt, \qquad \dot x^i = -N^i, \tag{1}$$

integrated along the fluid worldline from a growing-mode initial condition $\zeta_L(-\infty) = 0$ [19]. Every difference between Maldacena's ζ and the separate universe's δN_c is the divergence of the comoving shift, integrated along this worldline. At linear order, $\partial_i N^i = (\epsilon/c_s^2)\dot\zeta$ on super-

* houston@hubify.com; ORCID: 0009-0008-5616-5994

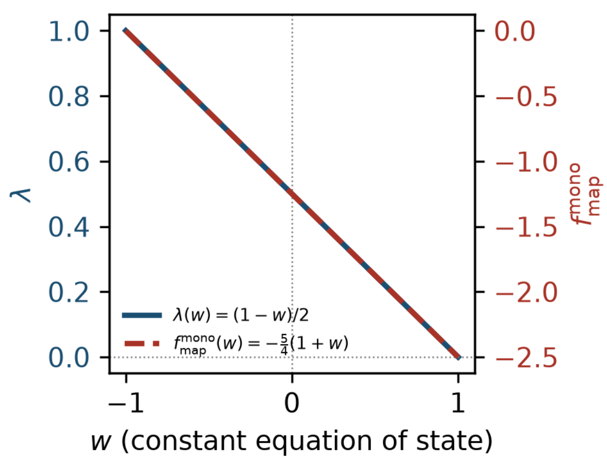

FIG. 1. The linear rescaling $\lambda(w) = (1-w)/2$ and the second-order map monopole $f^{\mathrm{mono}}_{\mathrm{map}}(w) = -\frac{5}{4}(1+w)$, Eq. (4), for constant equation of state $-1 \le w \le 1$ at $c_s = 1$. Both vanish at $w = -1$ (attractor limit); λ vanishes at $w = 1$ (kination), where δN_c loses linear response.

Hubble scales for any c_s [1, 2], so

$$\delta N_c = \lambda\,\zeta_L, \qquad \lambda = 1 - \frac{\langle \epsilon/c_s^2\rangle_\zeta}{3}, \qquad \langle X\rangle_\zeta \equiv \frac{\int X\,d\zeta_L}{\zeta_L(t_f)}, \tag{2}$$

exact for any single-field history with a growing-mode initial condition. This reduces to $\lambda = 1-\epsilon/3$ for constant ϵ, and to $\lambda_{\mathrm{USR}} = 1 + \epsilon_f/3 - \sqrt{\epsilon_s\epsilon_f}/3$ exactly for USR with $\epsilon = \epsilon_s(a/a_s)^{-6}$, $\zeta_L \propto a^3$. At second order, for constant ϵ and $c_s = 1$, solving the exact ADM constraints to $O(\zeta_L\zeta_S)$ and integrating Eq. (1) gives $f_{\delta N} = f^{\text{in-in}}/\lambda + f_{\mathrm{map}}$ with

$$f_{\mathrm{map}}(\epsilon,\mu) = -\frac{5\epsilon}{4}(1-\mu^2), \tag{3}$$

$\mu = \hat{k}_L\cdot\hat{k}_S$; every one of the five geometric contributions to f_{map} carries an explicit overall factor of ϵ [19]. In terms of a constant equation of state w ($\epsilon = \frac{3}{2}(1+w)$, $c_s = 1$),

$$\lambda = \frac{1-w}{2}, \qquad f_{\mathrm{map}} = -\frac{15}{8}(1+w)(1-\mu^2), \tag{4}$$

with monopole $f^{\mathrm{mono}}_{\mathrm{map}} = -\frac{5}{4}(1+w)$. Fig. 1 shows both across $-1 \le w \le 1$: $\lambda \to 1$, $f_{\mathrm{map}} \to 0$ at $w = -1$ (de Sitter, $\epsilon = 0$); and $\lambda \to 0$ at $w = 1$ (kination, $\epsilon = 3$), where δN_c ceases to be a usable linear-response variable and the in-in monopole itself vanishes.

III. THE CRITERION

Equations (2)–(3) give a single statement:

The isotropic separate universe (δN, with $N(\phi,\pi)$) reproduces the squeezed bispectrum of ζ iff the ζ-growth-weighted mean $\langle\epsilon/c_s^2\rangle_\zeta$ vanishes. When $\langle\epsilon/c_s^2\rangle_\zeta = O(1)$ – every constant-ϵ contraction with a growing ζ – the two differ at $O(1)$; when $\langle\epsilon/c_s^2\rangle_\zeta \ll 1$ (attractor: $\dot\zeta_L = 0$; USR: $\epsilon \ll 1$ throughout the growth) they agree to $O(\langle\epsilon/c_s^2\rangle_\zeta)$.

Equivalently, via the Hamiltonian constraint on comoving slices, the criterion is a statement about the long mode's comoving density contrast: the separate universe holds iff $\delta\rho_c/\rho \propto O(k^2/a^2H^2)\,\zeta_L$ (the gradient-expansion assumption of Lyth, Malik & Sasaki [13], following Salopek & Bond [14]), and fails at $O(1)$ exactly when the long mode's Bardeen potential grows as $\Psi_L = O((aH/k)^2)\,\zeta_L$, the textbook signature of a non-attractor contraction [15, 16]. Two logically distinct requirements are thus needed for δN to work in a non-attractor phase: (i) $N(\phi,\pi)$, to capture the growing mode as an independent initial datum [3]; (ii) $\langle\epsilon/c_s^2\rangle_\zeta \to 0$, so that the threading map is the identity. USR satisfies (ii) approximately because $\epsilon \ll 1$ while ζ grows – not because the comoving shift is small, which it is not: the shift itself is $O(1/k_L)$ in every non-attractor, USR included, but its coefficient $\epsilon\dot\zeta_L$ is what controls the $O(1)$ failure, and this coefficient is small in USR precisely when its growing-mode initial-data problem (i) is largest. Every inflationary test cited in Sec. I checked (i) with (ii) automatically small; none tested a regime with $\langle\epsilon/c_s^2\rangle_\zeta = O(1)$.

Table I summarizes the four validations (script: `separate_universe_failure_criterion_2026_09_04.py`, exact sympy). The dust case ($w = 0$) is the one already reported in Ref. [18]: the isotropic separate universe gives $f_{\delta N} = -5$ for every constant $\epsilon \in (1,3)$, while the in-in monopole varies with ϵ, so the two coincide only in the limit $\epsilon \to 0$. USR agrees because $\epsilon \ll 1$ throughout the growth of the long mode, exactly as the criterion predicts, and reduces to the $N(\phi,\pi)$ result of Ref. [3]. Attractor slow roll is the identity map, consistent with Maldacena's consistency relation [1]. Ekpyrotic contraction ($\epsilon \gg 3$) passes not because the background is expanding – it is not – but because the physical long mode sits on the constant-ζ solution, so $\langle\epsilon\rangle_\zeta = 0$; this reproduces the attractor behavior of single-field ekpyrosis found by Creminelli, Nicolis & Zaldarriaga [17]. The non-dominant, growing-Ψ ekpyrotic mode would fail by the same criterion, but never sets $\zeta_L(t_f)$ and so is not physically realized.

IV. WHAT IS NEW

Relative to Refs. [3, 4], this note identifies a second, independent failure mode of δN in non-attractors – the threading map $\delta N_c \neq \zeta$ – controlled by $\langle\epsilon/c_s^2\rangle_\zeta$ rather than by initial data, and shows it is $O(\epsilon)$ in USR (hence invisible to their test) and $O(1)$ in a matter-like contraction. Relative to Ref. [5], the attractor case reduces to $\Theta \equiv \epsilon\dot\zeta_L/(H\zeta_L) = 0$ and the identity map, so the two curvature variables coincide there and the only remaining question is observability; away from the attrac-

TABLE I. Validation of the criterion on four backgrounds (constant ϵ, $c_s = 1$ except USR). $\langle\epsilon\rangle_\zeta$ is the ζ-growth-weighted mean of ϵ/c_s^2; λ the linear rescaling; $f^{\text{in-in}}$ the in-in monopole; $f_{\delta N}$ the isotropic separate-universe value.

case	$\langle\epsilon\rangle_\zeta$	λ	$f^{\text{in-in}}$	$f_{\delta N}$	result
dust, $\epsilon = 3/2$	$3/2$	$1/2$	$-35/16 + \frac{15}{16}\mu^2$	-5	$O(1)$ fail, monopole gap $25/8$
USR, $\epsilon \propto a^{-6}$	$\ll 1$	$1 + O(\epsilon)$	$5/2$	$5/2$	agree to $O(\epsilon)$
attractor slow roll	0	1	$\frac{5}{12}(1 - n_s)$	same	identity map
ekpyrosis, dominant mode	0	1	attractor-like	same	passes (ζ on constant mode)

tor they already differ at $O(1)$ before any observability argument, by an explicit, time-dependent, non-local map along the fluid worldline. Relative to Ref. [6], this note supplies the input a conformal-Fermi-coordinate treatment of a genuine non-attractor would need: the $O(k^0)$ divergence of the long mode's comoving shift and the resulting displacement and dilation of the local frame. Relative to Refs. [7, 10], the agreement found there is explained as the $\epsilon \ll 1$ corner of the criterion rather than a coincidence of the USR transition. Relative to Refs. [8, 9], the coordinate-removal argument for the canonical single-field local $f_{\rm NL}$ is $O(\epsilon)$-controlled in inflation; whether an analogous physical-coordinate argument removes the $O(1)$ in-in monopole in a matter-like contraction is a distinct, open question, not addressed here. Relative to Refs. [11, 12], the mechanism and order differ: their breakdown is localized to a finite window of scales at a slow-roll-to-USR transition and traced to a momentum mismatch, while the criterion here applies to all super-Hubble scales at $O(k^0)$, at constant ϵ, with no transition required.

V. LIMITS

The second-order map, Eqs. (3)–(4), is exact for constant ϵ and $c_s = 1$ (the scalar-field realization of a given w); it is not the second-order map for a genuine fluid with $c_s^2 = w \neq 1$, which needs the c_s-dependent second-order ADM constraints – an open, bounded extension. The USR second-order statement in Sec. III is structural (every frozen second-order kernel is proportional to ϵ, and vanishes as $\epsilon \to 0$), not a re-solve with a time-dependent $\epsilon(t) = \epsilon_s(a/a_s)^{-6}$; a full time-dependent-$\epsilon$ second-order calculation was not attempted. The linear criterion, Eq. (2), is exact for any single-field history and any c_s, given a growing-mode initial condition.

REPRODUCIBILITY STATEMENT

The linear criterion, second-order map, general-w closed forms, and all four validations are computed by exact sympy in the committed script research/theory_audit/separate_universe_failure_criterion_2026_09_04.py (SHA-256 `21668ab6...`), with results in the companion `.json` (SHA-256 `4e2a49be...`) and derivation in the companion note of the same base name (`.md`); the second-order threading map is derived in threading_map_second_order_2026_09_04.{`md,py,json`}, and the in-in inputs in fnl_monopole_adjudication_2026_09_03.md. Figure 1 is generated by `make_fig_lambda_fmap.py` in this directory, transcribing the general-w closed forms with no new derivation. Manifest: reproducibility/manifests/experiments/lift2-separate-universe-failure-criterion.json. Local CPU, \$0, under 1 second total.

AI USAGE DISCLOSURE

The symbolic derivations underlying this note were performed by deterministic sympy scripts written and executed under the author's direction; every intermediate result is printed by the committed scripts and cross-checked against the frozen inputs before use. Claude (Anthropic) assisted with literature cross-checks, prose drafting, and LaTeX formatting under the author's direction; all physics content, equations, and numerical results were verified by the author against the committed script output.

[1] J. Maldacena, JHEP **05**, 013 (2003), arXiv:astro-ph/0210603.

[2] X. Chen, M. Huang, S. Kachru, G. Shiu, JCAP **0701**, 002 (2007), arXiv:hep-th/0605045.

[3] M. H. Namjoo, H. Firouzjahi, M. Sasaki, Europhys. Lett. **101**, 39001 (2013), arXiv:1210.3692.

[4] X. Chen, H. Firouzjahi, M. H. Namjoo, M. Sasaki, Europhys. Lett. **102**, 59001 (2013), arXiv:1301.5699.

[5] E. Pajer, F. Schmidt, M. Zaldarriaga, Phys. Rev. D **88**, 083502 (2013), arXiv:1305.0824.

[6] L. Dai, E. Pajer, F. Schmidt, JCAP **1511**, 043 (2015), arXiv:1504.00351.

[7] Y. Cai, X. Chen, M. H. Namjoo, M. Sasaki, D.-G. Wang, Z. Wang, JCAP **1805**, 012 (2018), arXiv:1712.09998.

[8] R. Bravo, S. Mooij, G. A. Palma, B. Pradenas, JCAP **1805**, 025 (2018), arXiv:1711.02680.

[9] R. Bravo, S. Mooij, G. A. Palma, B. Pradenas, JCAP **1805**, 024 (2018), arXiv:1711.05290.
[10] S. Passaglia, W. Hu, H. Motohashi, Phys. Rev. D **99**, 043536 (2019), arXiv:1812.08243.
[11] T. Artigas, S. Grain, V. Vennin, JCAP **02**, 001 (2022), arXiv:2110.11720.
[12] M. G. Jackson *et al.*, arXiv:2311.03281 (2023).
[13] D. H. Lyth, K. A. Malik, M. Sasaki, JCAP **0505**, 004 (2005), arXiv:astro-ph/0411220.
[14] D. S. Salopek, J. R. Bond, Phys. Rev. D **42**, 3936 (1990).
[15] D. Wands, Phys. Rev. D **60**, 023507 (1999), arXiv:gr-qc/9809062.
[16] F. Finelli, R. Brandenberger, Phys. Rev. D **65**, 103522 (2002), arXiv:hep-th/0112249.
[17] P. Creminelli, A. Nicolis, M. Zaldarriaga, Phys. Rev. D **71**, 063505 (2005), arXiv:hep-th/0411270.
[18] H. Golden, "$f_{\rm NL}$ monopole adjudication (2026-09-03)," https://github.com/Hubify-Projects/bigbounce/blob/main/research/theory_audit/fnl_monopole_adjudication_2026_09_03.md.
[19] H. Golden, "Threading map at second order (2026-09-04)," https://github.com/Hubify-Projects/bigbounce/blob/main/research/theory_audit/threading_map_second_order_2026_09_04.md.